

Phase Theory: Admissibility, Emergence, and the Reconstruction of Quantum and Relativistic Physics

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Abstract

We present a technical formulation of Phase Theory, a phase-primary framework in which physical reality is described by globally admissible phase configurations rather than particles, fields, or wavefunctions taken as primitives. We define the admissibility principle, formalize phase configuration space, and show how quantum mechanics, quantum field theory, and relativistic spacetime emerge as effective limits. Measurement is treated as admissibility restriction rather than collapse, and probability arises from unresolved phase distinguishability rather than intrinsic randomness. We identify regimes in which Phase Theory is empirically equivalent to standard formalisms and regimes—particularly at extreme coherence—where deviations are predicted. This article provides the mathematical and conceptual backbone underlying the accompanying Perspective.

1. Introduction

Foundational physics currently operates with a division of labor:

- Quantum mechanics supplies probabilistic dynamics without clear ontology.
- Quantum field theory supplies calculational power at the cost of renormalization and vacuum ambiguity.
- General relativity supplies geometric spacetime while resisting quantization.

Each framework is internally successful yet mutually strained. Interpretations proliferate not because the mathematics fails, but because primitive assumptions conflict.

Phase Theory proposes a single technical move: replace wavefunctions, fields, and spacetime as primitives with *phase* as the sole ontological substrate, subject to global admissibility constraints.

This article formalizes that move.

2. Phase Configuration Space

2.1 Phase as Primitive

We take phase to be a real, relational structure Φ defined without reference to spacetime points, particles, or fields. Formally, we consider a phase configuration space:

$$\mathcal{P} = \{\Phi \mid \Phi \text{ is globally consistent}\}$$

No Hilbert space is assumed at the fundamental level. Hilbert structures will arise as effective coordinate systems on admissible subsets of $\mathcal{P}$.

2.2 Admissibility

Not all phase configurations are physically realizable.

Definition (Admissibility).

A phase configuration Φ is *admissible* if it satisfies:

1. **Global consistency** — no contradictory phase relations.
2. **Topological coherence** — phase winding and defect structure are conserved.
3. **Bounded distinguishability** — finite resolution of phase distinctions.
4. **Saturation limits** — no unbounded phase gradients.

Let $\mathcal{A} \subset \mathcal{P}$ denote the admissible subset.

Inadmissible configurations are not dynamically forbidden—they are ontologically excluded.

3. Dynamics as Admissibility-Preserving Reconfiguration

Phase Theory does not postulate equations of motion as primitive laws.

Instead, evolution is defined as:

$$\Phi(t + \delta t) = \arg \min_{\Phi' \in \mathcal{A}} \mathcal{J}[\Phi' \mid \Phi(t)]$$

where $\mathcal{J}$ is a global consistency functional measuring admissibility cost.

This recovers:

- deterministic microscopic evolution,
- bounded macroscopic irreversibility,
- stability without imposed conservation laws.

4. Emergence of Quantum Mechanics

4.1 Discrete Sectors and Quantization

Quantization arises because admissible phase configurations partition into topologically distinct sectors. Continuous interpolation between sectors is inadmissible, yielding discrete spectra without postulated operators.

4.2 Effective Hilbert Space

In regimes of limited coherence and weak coupling, admissible phase submanifolds admit a linear representation:

$$\Phi \rightarrow |\psi\rangle \in \mathcal{H}_{\text{eff}}$$

This mapping is:

- approximate,
- non-unique,
- regime-dependent.

Hilbert space is *descriptive*, not ontological.

4.3 Schrödinger Dynamics as a Limit

The Schrödinger equation emerges as the linearized flow of Φ near stable admissible trajectories:

$$\hbar \partial/\partial t |\psi\rangle = \hat{H}_{\text{eff}} |\psi\rangle$$

with $\hat{H}_{\text{eff}}$ encoding local admissibility geometry.

5. Measurement Without Collapse

Measurement corresponds to restriction of admissible phase structure due to coupling with an environment or apparatus.

- No discontinuous collapse occurs.
- Outcome selection precedes decoherence.
- Irreversibility arises only when admissibility saturates.

Formally, measurement maps:

$$\mathcal{A} \rightarrow \mathcal{A} \subset \mathcal{A}$$

Collapse postulates are unnecessary.

6. Probability as Phase Statistics

Phase Theory does not posit intrinsic randomness.

Probabilities arise from:

- coarse-graining over unresolved admissible configurations,
- relative basin volumes in phase space.

Born-rule statistics appear as:

$$P_i \sim \mu(\mathcal{B}_i)$$

where $\mathcal{B}_i$ are admissible basins and μ is a phase-consistent measure.

7. Emergence of Spacetime and Relativity

Large-scale phase ordering induces an effective metric structure:

$$g_{\mu\nu} \sim \partial_\mu \Phi \partial_\nu \Phi$$

Lorentz invariance arises as a symmetry of admissible phase relations, not as a background assumption.

Gravitational phenomena correspond to phase refraction, not force mediation.

8. Relation to Quantum Field Theory

Quantum field theory appears as:

- a coarse-grained description of collective phase modes,
- valid when admissibility constraints are weakly coupled.

Renormalization reflects loss of phase resolution, not physical divergence.

9. Experimental Deviations and Falsifiability

Phase Theory predicts deviations from standard formalisms in regimes of:

- extreme coherence,
- admissibility saturation,
- engineered phase geometry.

Examples include:

- non-Carnot entropy flow,
- geometry-dependent interaction strengths,

- partial reversibility of measurement.

These effects are experimentally accessible with current photonic and coherent-matter platforms.

10. Consistency and Limits

Phase Theory is:

- internally consistent,
- ontologically minimal,
- empirically constrained.

It does not claim:

- completeness beyond admissibility,
- immunity to falsification,
- reduction of all phenomena.

Its strength lies in replacing contradiction with constraint.

11. Discussion

Phase Theory reframes foundational physics by shifting attention from entities to consistency structure. It does not add degrees of freedom or speculative dimensions; it removes misplaced primitives.

Whether it ultimately succeeds is an experimental question. But it offers a technically precise alternative to collapse, intrinsic randomness, and interpretational proliferation—without sacrificing established results.

12. Conclusion

We have provided a technical formulation of Phase Theory suitable for engagement by the broader physics community. By grounding physics in admissible phase structure, quantum mechanics, field theory, and relativity emerge as effective limits of a single coherent framework. Measurement, probability, and spacetime are no longer fundamental mysteries but structural consequences.

"Phase Theory does not claim to end physics. It claims to clean its foundations."

Companion

to:

"Phase Theory: Rethinking Physical Foundations Through Phase Primacy"